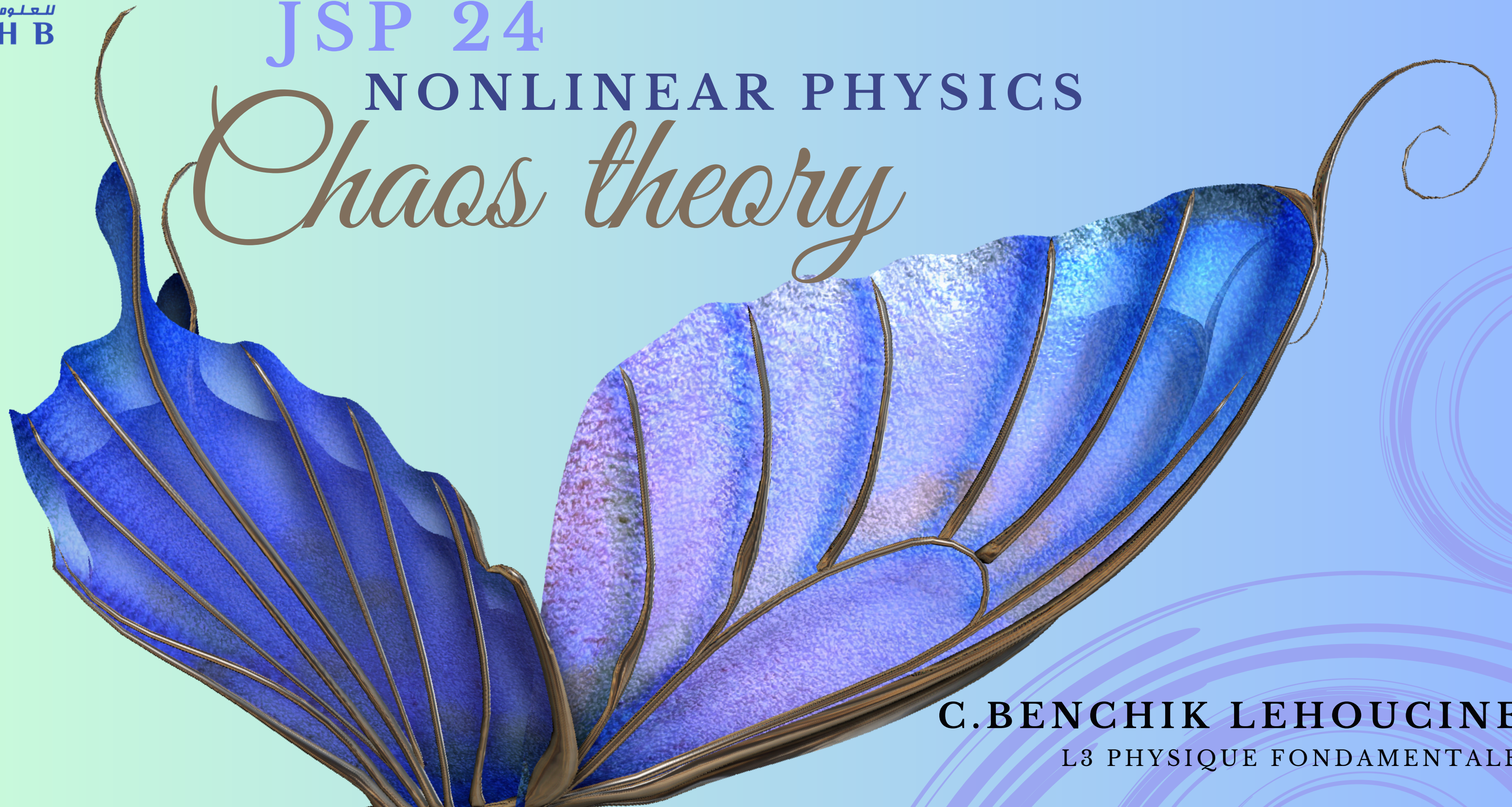


JSP 24

NONLINEAR PHYSICS

Chaos theory



C.BENCHIK LEHOUCINE
L3 PHYSIQUE FONDAMENTALE

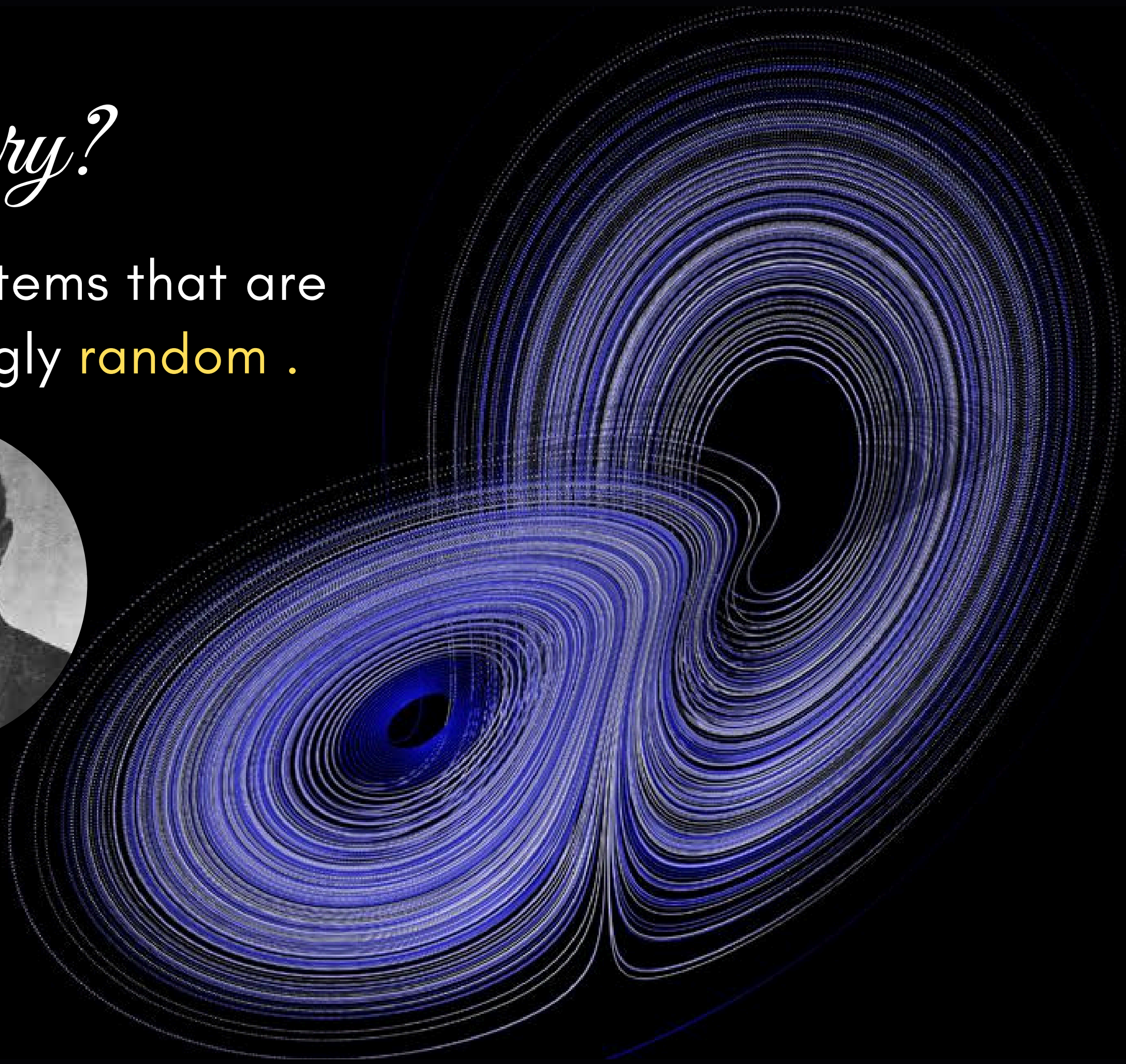
What is Chaos theory?

chaos theory deals with systems that are
unpredictable and seemingly **random** .

Henri Poincaré



Edward Lorenz



The Double Pendulum:

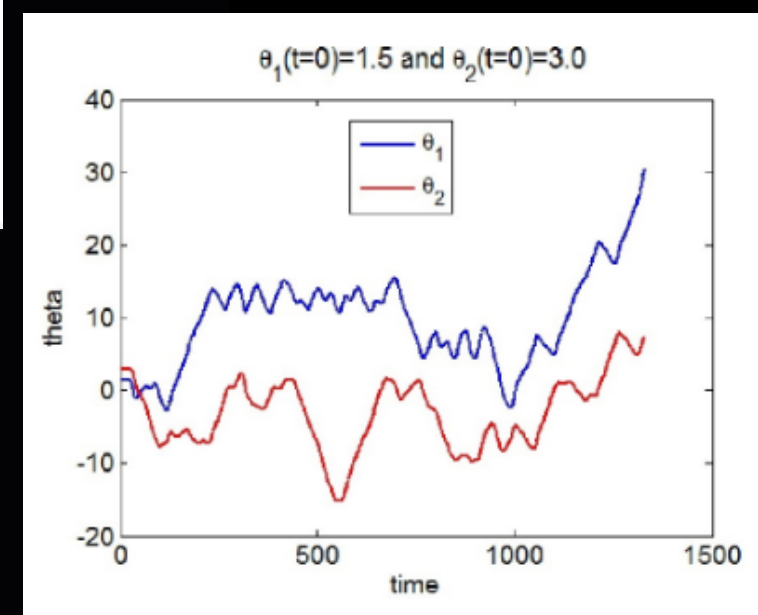
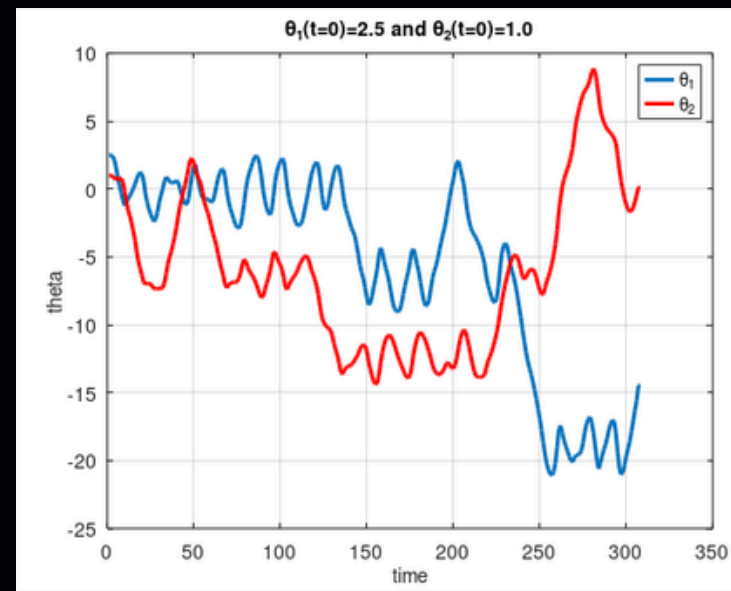
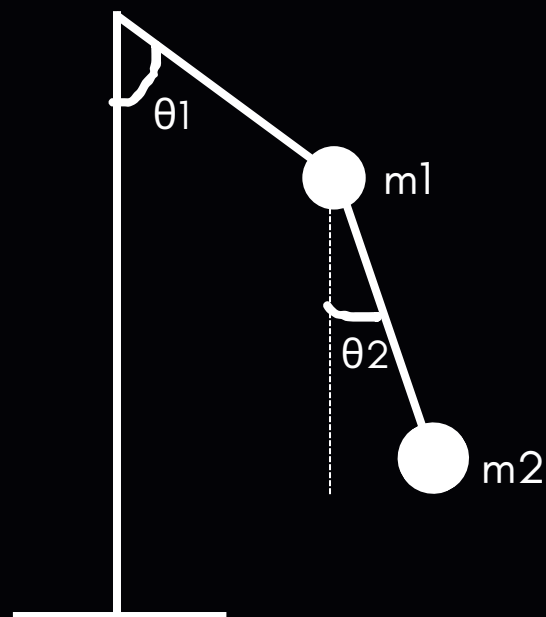
The **Lagrangian** is:

$$L = \frac{1}{2}(m_1 + m_2)l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2l_2^2\dot{\theta}_2^2 + m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) + (m_1 + m_2)gl_1\cos(\theta_1) + m_2gl_2\cos(\theta_2)$$

The Equations of **motion**:

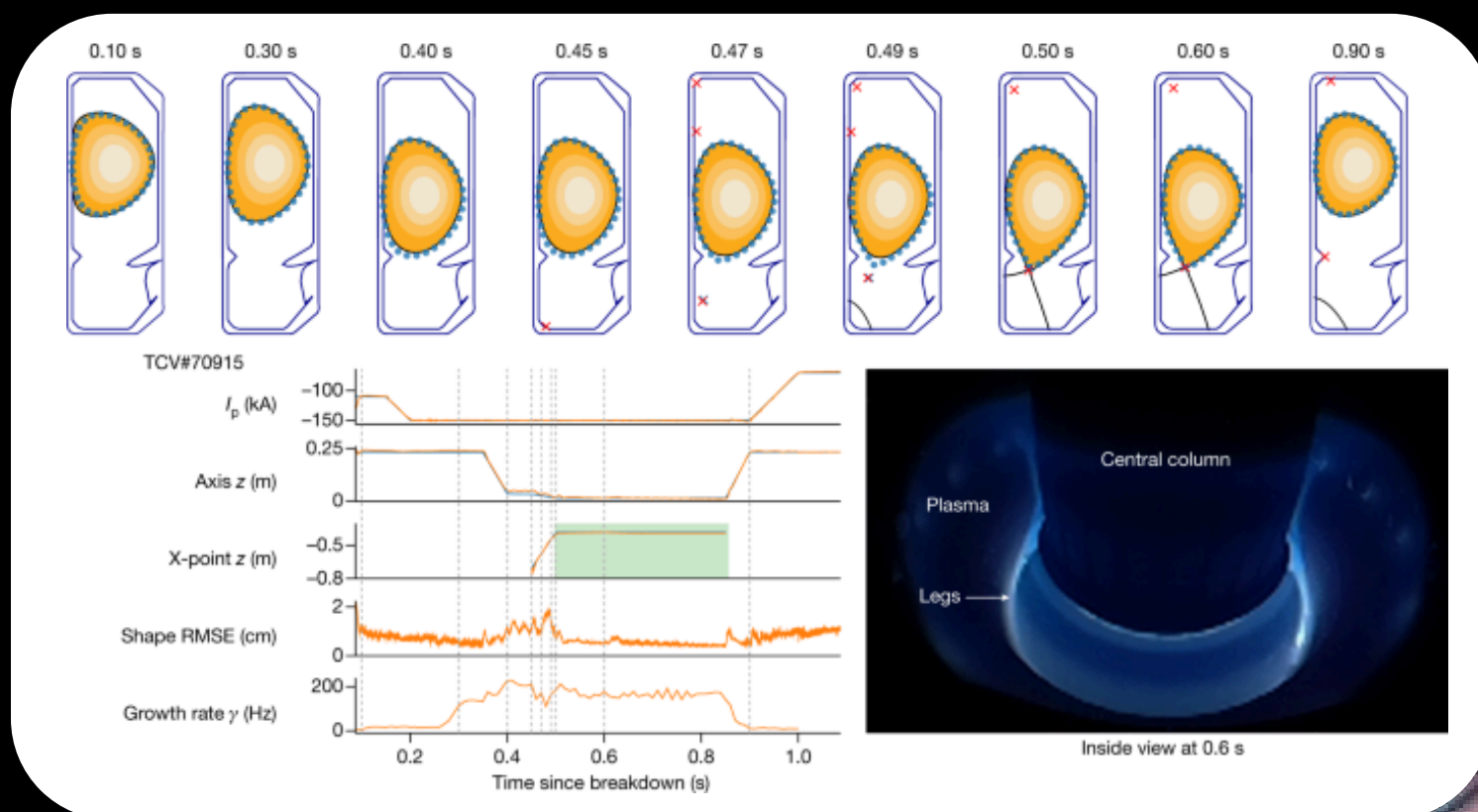
$$(1) (m_1 + m_2)l_1\ddot{\theta}_1 + m_2l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) + m_2l_2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) + (m_1 + m_2)g\sin(\theta_1) = 0$$

$$(2) l_1\ddot{\theta}_1\cos(\theta_1 - \theta_2) + l_2\ddot{\theta}_2 - l_1\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + g\sin(\theta_2) = 0$$

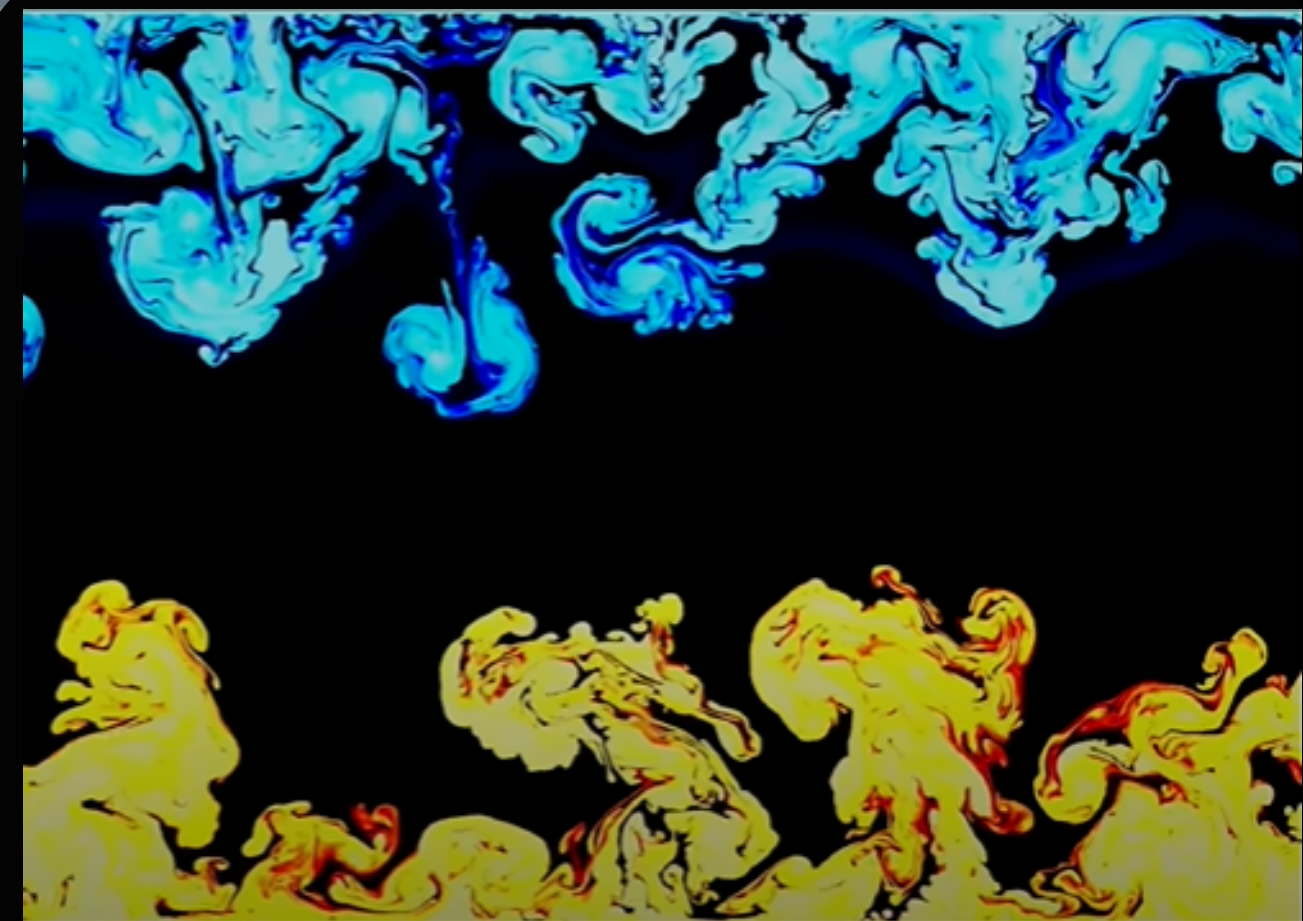


Chaos control application:

Chaos control in **nuclear fusion plasma**:



MAGNETIC CONTROL OF TOKAMAK PLASMAS THROUGH DEEP
REINFORCEMENT LEARNING | NATURE



Chaos control in **fluid**: